

SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

ANALOG AND DIGITAL ELECTRONICS

MODEL QUESTION PAPER – SET-1

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

Module Outcome Cognitive level

1	With respect to amount of doping arrange the regions of BJT	M 1.01	R
2	List down the different modes of operation of BJT	M1.02	R
3	Based on the working principle list the classification MOSFET	M2.02	U
4	Draw the symbol of N channel and P channel FET	M2.01	U
5	Write the truth table of a JK flipflop	M3.04	R
6	What is meant by Multiplexer	M3.02	U
7	List down the different process involved in analog to digital conversion.	M4.01	U
8	List down the classification of analog to digital convertor	M4.02	U
9	Draw the symbol of NPN and PNP transistor	M1.01	R

PART B

II. Answer any Eight questions from the following

(8 x 3= 24 Marks)

Module Outcome Cognitive level

1	Derive relationship between α and β of BJT	M 1.02	A
2	Differentiate between BJT and FET	M2.04	U
3	Design and implement full adder circuit using gates	M3.02	A
4	Design and implement 2*1 MUX using gates	M3.02	A
5	With circuit diagram explain the working of SR flipflop	M3.04	R
6	With circuit diagram explain the working of UJT relaxation oscillator	M2.03	U
7	Differentiate between CE CC CB configurations	M1.03	U

8	Briefly explain about quantisation	M4.01	R
9	Write short notes classification of analog to digital convertors	M4.02	R
10	Draw the structure and equivalent circuit of UJT	M2.03	U

PART C

III. Answer all questions from the following

(6x 7 = 42 Marks)

Module Outcome Cognitive level

1	Explain input and output characteristics of a common base configuration	M1.03	U
OR			
2	Explain amplification process of a common emitter configuration	M1.04	U
3	Explain the characteristics of a JFET	M2.02	U
OR			
4	Explain the input and output characteristics of a N channel depletion MOSFET	M2.03	U
5	With neat diagram explain the working of carry lookahead adder	M3.03	A
OR			
6	Design a 4bit synchronous up counter using JK flipflop	M3.04	A
7	With Diagram explain SISO and PISO shift registers	M3.05	U
OR			
8	Design a MOD 10 asynchronous counter using J K flipflop	M3.04	A
9	Explain the working successive approximation type ADC	M 4.03	R
OR			
10	Explain the working of R 2 R type DAC	M4.02	R
11	Using 4-bit adder design a 4-bit BCD adder	M3.03	A
OR			
12	Design 2 input OR, NAND, EXOR using 2*1 MUX	M3.01	A

SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

ANALOG AND DIGITAL ELECTRONICS

MODEL QUESTION PAPER – SET-2

Time: 3 hours

Maximum Marks: 75

PART A

IV. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	With respect to area of cross section arrange the regions of BJT	M 1.01	R
2	List down the different configurations of BJT	M1.02	R
3	Based on the working principle list the classification JFET	M2.02	U
4	Draw the symbol of N channel and P channel MOSFET	M2.01	U
5	Write the truth table of a SR flipflop	M3.04	R
6	What is meant by Demultiplexer	M3.02	U
7	List down the different process involved in analog to digital conversion.	M4.01	U
8	List down the classification of digital to analog convertors	M4.02	U
9	Draw the symbol of NPN and PNP transistor	M1.01	R

PART B

V. Answer any Eight questions from the following

(8 x 3= 24 Marks)

		Module Outcome	Cognitive level
1	Derive relationship between α and β of BJT	M 1.02	A
2	Differentiate between BJT and FET	M2.04	U
3	Design and implement full subtractor circuit using gates	M3.02	A
4	Design and implement 2*1 MUX NAND gates	M3.02	A
5	With circuit diagram explain the working of D flipflop	M3.04	R
6	With circuit diagram explain the working of UJT relaxation oscillator	M2.03	U

7	Differentiate between CE CC CB configurations	M1.03	U
8	Briefly explain about encoding in analog to digital conversion	M4.01	R
9	Write short notes classification of digital to analog convertors	M4.02	R
10	Define intrinsic standoff ratio of UJT	M2.03	U

PART C

VI. Answer all questions from the following (6x 7 = 42 Marks)

Module Outcome Cognitive level

1	Explain input and output characteristics of a common collector configuration	M1.03	U
OR			
2	Explain amplification process of a common emitter configuration	M1.04	U
3	Explain the characteristics of a MOSFET	M2.02	U
OR			
4	Explain the input and output characteristics of a JFET	M2.03	U
5	With neat diagram explain the working of carry lookahead adder	M3.03	A
OR			
6	Design a 4bit synchronous down counter using JK flipflop	M3.04	A
7	With Diagram explain SISO and PISO shift registers	M3.05	U
OR			
8	Design a MOD 8 asynchronous counter using J K flipflop	M3.04	A
9	Explain the working counter type ADC	M 4.03	R
OR			
10	Explain the working of weighted R type DAC	M4.02	R
11	Using full adder adder design a 4-bit adder	M3.03	A
OR			
12	Design 2 input OR, NAND, EXR using 2*1 MUX	M3.01	A